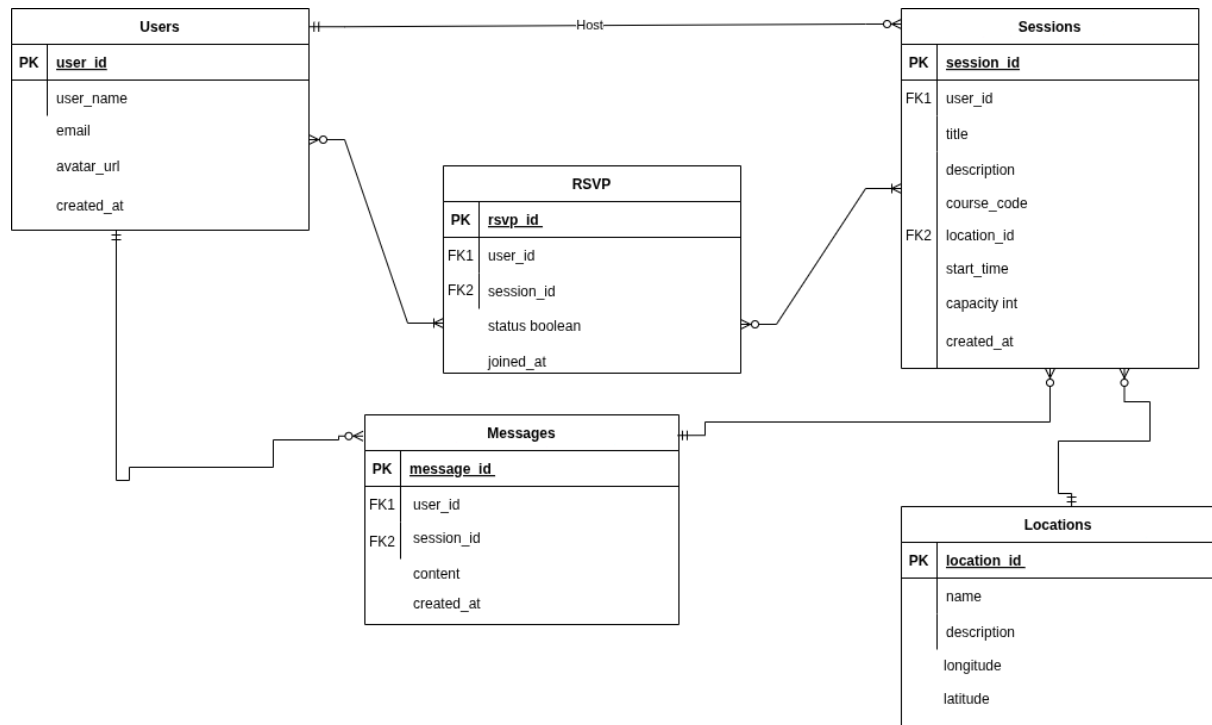


Database Design for StudyBuddy

Entity Relationship Diagram



Relationship	Type	Participation	FK Location	Constraints
User → Session (hosts)	1:N	User (0..N), Session (1..1)	sessions.host_id	NOT NULL FK
User ↔ Session (attends)	M:N	Optional both sides	rsvps.user_id, rsvps.session_id	UNIQUE(user_id, session_id)
Session → Message	1:N	Session (0..N), Message (1..1)	messages.session_id	NOT NULL FK
User → Message	1:N	User (0..N), Message (1..1)	messages.user_id	NOT NULL FK
Session → Location	N:1	Session (1..1), Location (0..N)	sessions.location_id	NOT NULL FK

Tables & Constraints

USERS

Column	Type	Constraints
id	PK	PRIMARY KEY
name	TEXT	NOT NULL
email	TEXT	UNIQUE, NOT NULL
avatar_url	TEXT	NULL
created_at	DATETIME	DEFAULT NOW()

LOCATION

Column	Type	Constraints
id	PK	PRIMARY KEY
name	TEXT	NOT NULL
description	TEXT	NULL
latitude	DECIMAL	NULL
longitude	DECIMAL	NULL

SESSION

Column	Type	Constraints
id	PK	PRIMARY KEY
title	TEXT	NOT NULL
description	TEXT	NULL
course_code	TEXT	NULL
location_id	FK	REFERENCES locations(id), NOT NULL
host_id	FK	REFERENCES users(id), NOT NULL
start_time	DATETIME	NOT NULL
capacity	INT	CHECK (capacity > 0)
created_at	DATETIME	DEFAULT NOW()

RSVP (JOINT TABLE)

Column	Type	Constraints
id	PK	PRIMARY KEY
user_id	FK	REFERENCES users(id), NOT NULL
session_id	FK	REFERENCES sessions(id), NOT NULL
status	TEXT	CHECK ('attending','waitlist','cancelled')
joined_at	DATETIME	DEFAULT NOW()
UNIQUE(user_id, session_id)	—	Prevent duplicates

MESSAGES

Column	Type	Constraints
id	PK	PRIMARY KEY
session_id	FK	REFERENCES sessions(id), NOT NULL
user_id	FK	REFERENCES users(id), NOT NULL
content	TEXT	NOT NULL
created_at	DATETIME	DEFAULT NOW()